

Summer Bootcamp – Session 3 Notes

Optimization

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Purpose of the Session

Optimization turns a modeling objective into a decision. This session develops the conditions used to recognize optima, the role of convexity in obtaining global guarantees, and the algorithms used to solve smooth and constrained problems. The final sections apply this toolkit to portfolio allocation.

The central logical distinction is the following:

derivatives generate candidates; curvature, convexity, feasibility, and duality determine whether those candidates are optimal.

1 Formulating an Optimization Problem

A general constrained optimization problem has the form

$$\min_{x \in \mathbb{R}^d} f(x) \quad \text{subject to} \quad h(x) = 0, \quad g(x) \leq 0.$$

Here x is the decision variable, f is the objective function, and

$$\mathcal{F} = \{x \in \mathbb{R}^d : h(x) = 0, g(x) \leq 0\}$$

is the feasible set.

Definition 1.1 (Local and global minimizers). A feasible point x^* is a local minimizer if it has no better feasible competitor in some neighborhood. It is a global minimizer if

$$f(x^*) \leq f(x) \quad \text{for every } x \in \mathcal{F}.$$

Three recurring financial examples are

$$\min_{\theta} \frac{1}{2} \|A\theta - b\|_2^2,$$

$$\min_w \frac{1}{2} w^\top \Sigma w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1,$$

and

$$\min_w \frac{1}{2} w^\top \Sigma w - \gamma \mu^\top w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad w \geq 0.$$

Example. Fitting $y_i \approx a + bt_i$ by least squares means choosing $x = (a, b)^\top$ and solving

$$\min_{a,b} \frac{1}{2} \sum_{i=1}^n (a + bt_i - y_i)^2 = \min_x \frac{1}{2} \|Ax - y\|_2^2,$$

with $A_i = (1, t_i)$. There is no constraint.

Example. If R contains asset returns and b contains benchmark returns, a fully invested long-only tracking portfolio solves

$$\min_{w \in \mathbb{R}^d} \frac{1}{2} \|Rw - b\|_2^2 \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad w \geq 0.$$

The decision variable is w and the feasible set is the long-only simplex.

2 Unconstrained Optimality

2.1 First-order condition

Proposition 2.1: First-order necessary condition

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be differentiable. If x^* is an interior local minimum, then

$$\nabla f(x^*) = 0.$$

Such a point is called a critical or stationary point.

Proof. For every direction $v \in \mathbb{R}^d$, define $\phi(t) = f(x^* + tv)$. Since $t = 0$ is a local minimum of ϕ ,

$$0 = \phi'(0) = \nabla f(x^*)^\top v.$$

This holds for every v , so $\nabla f(x^*) = 0$. □

Necessary does not mean sufficient

A stationary point can be a minimum, a maximum, or a saddle point. The first-order equation only generates candidates; classification requires curvature or another argument.

2.2 Second-order information

For a twice differentiable function, Taylor expansion at a critical point gives

$$f(x^* + h) = f(x^*) + \frac{1}{2} h^\top \nabla^2 f(x^*) h + o(\|h\|^2).$$

Proposition 2.2: Second-order classification

Let x^* satisfy $\nabla f(x^*) = 0$.

- If $\nabla^2 f(x^*) \succ 0$, then x^* is a strict local minimum.
- If $\nabla^2 f(x^*) \prec 0$, then x^* is a strict local maximum.
- If $\nabla^2 f(x^*)$ is indefinite, then x^* is a saddle point.
- A semidefinite Hessian can be inconclusive.

Reminder: positivity of symmetric matrices

For $H = H^\top$, the sign of the quadratic form $z^\top H z$ determines its definiteness:

$$\begin{aligned} H \succ 0 &\iff z^\top H z > 0 \quad \text{for all } z \neq 0, \\ H \succeq 0 &\iff z^\top H z \geq 0 \quad \text{for all } z, \\ H \prec 0 &\iff z^\top H z < 0 \quad \text{for all } z \neq 0. \end{aligned}$$

If the quadratic form takes both signs, H is indefinite. For symmetric matrices, these conditions are equivalent to the corresponding sign conditions on every eigenvalue.

For a 2×2 symmetric matrix

$$H = \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

the easy determinant tests are

$H \succ 0$	$\iff$	$a > 0$ and $ac - b^2 > 0$,
$H \prec 0$	$\iff$	$a < 0$ and $ac - b^2 > 0$,
H indefinite	$\iff$	$ac - b^2 < 0$.

In any dimension, Sylvester's criterion states that $H \succ 0$ exactly when all leading principal minors are positive.

Example. For $f(x) = x^3 - 3x^2 + 2$, the stationary points are 0 and 2. Since $f''(0) = -6$ and $f''(2) = 6$, they are respectively a local maximum and a local minimum. For

$$q(x, y) = x^2 + xy + y^2 - 4x - 2y,$$

the stationary point is $(2, 0)$ and

$$H_q = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \succ 0,$$

so it is the unique global minimum. Finally, $x^2 - y^2$ has an indefinite Hessian at the origin and therefore a saddle point there.

Example. For

$$g_1(x, y) = 3x^2 + 2xy + 2y^2, \quad g_2(x, y) = -x^2 - xy - y^2,$$

we have

$$H_{g_1} = \begin{pmatrix} 6 & 2 \\ 2 & 4 \end{pmatrix} \succ 0, \quad H_{g_2} = \begin{pmatrix} -2 & -1 \\ -1 & -2 \end{pmatrix} \prec 0.$$

Thus the origin is a strict minimum of g_1 and a strict maximum of g_2 .

The functions $x \mapsto x^4$ and $x \mapsto -x^4$ both have zero first and second derivatives at the origin, yet the origin is respectively a strict minimum and a strict maximum. Higher-order information is then necessary.

2.3 Quadratic objectives and least squares

Consider

$$f(x) = \frac{1}{2}x^\top Qx + c^\top x + r, \quad Q = Q^\top.$$

Then

$$\nabla f(x) = Qx + c, \quad \nabla^2 f(x) = Q.$$

Proposition 2.3: Minimizer of a positive-definite quadratic

If $Q \succ 0$, the function has the unique global minimizer

$$x^\star = -Q^{-1}c.$$

For least squares,

$$f(x) = \frac{1}{2} \|Ax - b\|_2^2, \quad \nabla f(x) = A^\top(Ax - b).$$

The criticality condition produces the normal equations

$$A^\top Ax = A^\top b.$$

If A has full column rank, then $A^\top A \succ 0$ and the solution is unique.

3 Convexity and Global Optimality

Definition 3.1 (Convex set). A set $C \subset \mathbb{R}^d$ is convex if

$$x, y \in C, t \in [0, 1] \implies tx + (1 - t)y \in C.$$

Definition 3.2 (Convex function). A function $f : C \rightarrow \mathbb{R}$ is convex if

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y)$$

for all $x, y \in C$ and $t \in [0, 1]$.

Proposition 3.1: First-order characterization of convexity

Let f be differentiable on an open convex set. Then f is convex if and only if

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) \quad \text{for all } x, y.$$

Thus the tangent affine function is a global lower bound.

Proposition 3.2: Criticality is sufficient under convexity

If f is differentiable and convex and

$$\nabla f(x^*) = 0,$$

then x^* is a global minimizer.

Indeed, the first-order inequality gives $f(y) \geq f(x^*)$ for every y .

Proposition 3.3: Second-order characterization of convexity

Let f be twice differentiable on an open convex set. Then

$$f \text{ is convex} \iff \nabla^2 f(x) \succeq 0 \quad \text{for every } x.$$

Example. The functions

$$e^{a^\top x}, \quad \log(1 + e^x), \quad x^2 - xy + y^2$$

are convex because their Hessians are respectively

$$e^{a^\top x} aa^\top \succeq 0, \quad \frac{e^x}{(1 + e^x)^2} > 0, \quad \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \succ 0.$$

Example. For $p(x, y) = x^4 + e^y$,

$$\nabla^2 p(x, y) = \begin{pmatrix} 12x^2 & 0 \\ 0 & e^y \end{pmatrix} \succeq 0,$$

so p is convex. For $q(x, y) = x^2 - y^2$, the Hessian $\text{diag}(2, -2)$ is indefinite, so q is not convex.

Definition 3.3 (Strong convexity and smoothness). A twice differentiable function is μ -strongly convex and L -smooth when

$$\mu I \preceq \nabla^2 f(x) \preceq LI$$

throughout its domain. The ratio $\kappa = L/\mu$ is the condition number.

Large κ corresponds to very different curvatures in different directions. This geometry typically slows first-order algorithms.

For example, the Hessian of $(x^2 + 100y^2)/2$ has eigenvalues 1 and 100, so $(\mu, L, \kappa) = (1, 100, 100)$. As a second calculation, the Hessian of $2x^2 + 32y^2$ is $\text{diag}(4, 64)$, hence

$$\mu = 4, \quad L = 64, \quad \kappa = 16.$$

4 Algorithms for Smooth Optimization

4.1 Gradient descent

Gradient descent uses the iteration

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k).$$

The negative gradient is the direction of steepest local decrease in the Euclidean geometry.

Proposition 4.1: Descent lemma

If ∇f is L -Lipschitz, then

$$f(x - \alpha \nabla f(x)) \leq f(x) - \alpha \left(1 - \frac{L\alpha}{2}\right) \|\nabla f(x)\|_2^2.$$

Consequently, every fixed step satisfying

$$0 < \alpha < \frac{2}{L}$$

decreases the objective away from stationary points.

A standard conservative choice is $\alpha = 1/L$. When L is unknown, backtracking line search reduces the step until sufficient decrease is observed.

Example. For $f(x) = \frac{1}{2}(x - 3)^2$, $x_0 = 0$, and $\alpha = 0.4$,

$$x_{k+1} = 0.6x_k + 1.2, \quad (x_1, x_2, x_3) = (1.2, 1.92, 2.352).$$

For the companion problem $g(x) = \frac{1}{2}(x + 2)^2$, $x_0 = 2$, and $\alpha = 0.25$,

$$x_{k+1} = 0.75x_k - 0.5, \quad (x_1, x_2, x_3) = (1, 0.25, -0.3125).$$

Proposition 4.2: Linear convergence under strong convexity

If f is μ -strongly convex and L -smooth, gradient descent with $\alpha = 1/L$ satisfies

$$f(x_k) - f(x^*) \leq \left(1 - \frac{\mu}{L}\right)^k (f(x_0) - f(x^*)).$$

4.2 Newton's method

The second-order approximation

$$f(x + p) \approx f(x) + \nabla f(x)^\top p + \frac{1}{2} p^\top \nabla^2 f(x) p$$

is minimized by solving

$$\nabla^2 f(x_k) p_k = -\nabla f(x_k), \quad x_{k+1} = x_k + p_k.$$

Implementation rule

Solve the linear system for p_k ; do not explicitly form the inverse Hessian. Globalization may require damping or a line search, especially when the Hessian is indefinite or the initial point is far from the solution.

Example. For $f(x) = e^x - 3x$, Newton's update is

$$x_{k+1} = x_k - 1 + 3e^{-x_k}.$$

Starting from $x_0 = 1$ gives $x_1 \simeq 1.10364$ and $x_2 \simeq 1.09862$, close to $\log 3$. For the companion problem $g(x) = e^x - 2x$ from $x_0 = 0.5$,

$$x_{k+1} = x_k - 1 + 2e^{-x_k}, \quad \boxed{x_1 \simeq 0.71306, \quad x_2 \simeq 0.69334,}$$

close to $\log 2$.

4.3 Projected gradient descent

We now solve the convex constrained problem

$$\min_{x \in C} f(x),$$

where f is differentiable and convex and C is nonempty, closed, and convex.

Proposition 4.3: Euclidean projection onto a convex set

For every $z \in \mathbb{R}^d$, the projection

$$\text{proj}_C(z) = \arg \min_{u \in C} \|u - z\|_2$$

exists and is unique.

Projected gradient descent alternates an unconstrained step and a projection:

$$\boxed{x_{k+1} = \text{proj}_C(x_k - \alpha_k \nabla f(x_k)).}$$

5 Equality Constraints and Lagrange Multipliers

Consider

$$\min_x f(x) \quad \text{s.t.} \quad h(x) = 0.$$

At a regular feasible point, first-order feasible directions satisfy $\nabla h(x)^\top v = 0$. At an optimum, $\nabla f(x^*)$ must also be orthogonal to all feasible directions.

Proposition 5.1: Lagrange first-order conditions

Define

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^\top h(x).$$

Under a regularity condition, a constrained local optimum satisfies

$$\boxed{\nabla_x \mathcal{L}(x^*, \lambda^*) = 0, \quad h(x^*) = 0.}$$

These equations are necessary in general, not sufficient. Convexity or a second-order constrained argument is required to classify candidates.

Example. For

$$\min_{x,y} x^2 + y^2 \quad \text{s.t.} \quad x + y = 1,$$

the equations $2x + \lambda = 0$, $2y + \lambda = 0$, and $x + y = 1$ give

$$\boxed{x^* = y^* = \frac{1}{2}, \quad \lambda^* = -1.}$$

Example. For

$$\min_{x,y} x^2 + y^2 \quad \text{s.t.} \quad 2x - y = 3,$$

the Lagrangian is $x^2 + y^2 + \lambda(2x - y - 3)$. The first-order equations give $x = -\lambda$, $y = \lambda/2$, and therefore

$$\lambda^* = -\frac{6}{5}, \quad (x^*, y^*) = \left(\frac{6}{5}, -\frac{3}{5}\right).$$

For several equality constraints $h = (h_1, \dots, h_m)$, a standard regularity condition is LICQ.

Proposition 5.2: Linear independence constraint qualification

LICQ holds at x^* when

$$\nabla h_1(x^*), \dots, \nabla h_m(x^*) \text{ are linearly independent.}$$

Why regularity matters. For $\min_x x$ subject to $x^2 = 0$, the only feasible point is 0, but $\partial_x(x + \lambda x^2)|_{x=0} = 1$ for every λ . Likewise, for

$$\min_{x,y} x + y \quad \text{s.t.} \quad x^2 + y^2 = 0,$$

the only feasible point is $(0, 0)$ while $\nabla f(0, 0) = (1, 1)^\top$ and $\nabla h(0, 0) = 0$. In both cases the constraint gradient vanishes, LICQ fails, and no multiplier can satisfy stationarity.

5.1 Duality

The multiplier λ_i can be interpreted as the marginal price of violating constraint i . The dual function is

$$q(\lambda) = \inf_x \mathcal{L}(x, \lambda).$$

It is concave because it is the pointwise infimum of affine functions of λ .

Proposition 5.3: Weak duality

Let

$$p^* = \inf_{h(x)=0} f(x), \quad d^* = \sup_{\lambda} q(\lambda).$$

Then, without requiring convexity,

$$d^* \leq p^*.$$

For every feasible x and every λ ,

$$f(x) - q(\lambda) \geq 0.$$

The latter quantity is the primal–dual gap.

A small primal–dual gap gives a computable certificate of near-optimality.

Proposition 5.4: Strong duality and saddle points

Strong duality means

$$d^* = p^*.$$

For convex problems, affine equality constraints and a suitable constraint qualification commonly ensure this equality. If optimal solutions exist, then

$$\min_x \max_{\lambda} \mathcal{L}(x, \lambda) = \max_{\lambda} \min_x \mathcal{L}(x, \lambda).$$

The optimal pair is a saddle point:

$$\mathcal{L}(x^*, \lambda) \leq \mathcal{L}(x^*, \lambda^*) \leq \mathcal{L}(x, \lambda^*).$$

5.2 Primal–dual gradient method

For a differentiable Lagrangian, the direct descent–ascent iteration is

$$\begin{aligned} x_{k+1} &= x_k - \alpha_k \nabla_x \mathcal{L}(x_k, \lambda_k), \\ \lambda_{k+1} &= \lambda_k + \beta_k \nabla_\lambda \mathcal{L}(x_k, \lambda_k). \end{aligned}$$

Since $\nabla_\lambda \mathcal{L}(x, \lambda) = h(x)$,

$$\lambda_{k+1} = \lambda_k + \beta_k h(x_k).$$

The primal step decreases the Lagrangian with respect to the decision, while the dual step raises the price of constraint violations.

Possible oscillations

Simultaneous gradient descent–ascent can rotate around a saddle point rather than converge. Damping, extragradient methods, and augmented Lagrangians can improve stability.

6 Inequality Constraints and KKT Conditions

Consider

$$\min_x f(x) \quad \text{s.t.} \quad h(x) = 0, \quad g(x) \leq 0,$$

with

$$\mathcal{L}(x, \nu, \lambda) = f(x) + \nu^\top h(x) + \lambda^\top g(x).$$

Proposition 6.1: Karush–Kuhn–Tucker necessary conditions

Under a suitable constraint qualification, a local optimum admits multipliers such that

$$\begin{aligned} \nabla_x \mathcal{L}(x^*, \nu^*, \lambda^*) &= 0 && \text{stationarity,} \\ h(x^*) &= 0, \quad g(x^*) \leq 0 && \text{primal feasibility,} \\ \lambda^* &\geq 0 && \text{dual feasibility,} \\ \lambda_i^* g_i(x^*) &= 0 && \text{complementary slackness.} \end{aligned}$$

Complementary slackness separates active and inactive constraints:

- if $g_i(x^*) < 0$, then $\lambda_i^* = 0$;
- if $\lambda_i^* > 0$, then $g_i(x^*) = 0$;
- an active constraint may still have a zero multiplier.

Example. For

$$\min_{x,y} (x-1)^2 + (y-2)^2 \quad \text{s.t.} \quad x+y \leq 1,$$

the unconstrained minimizer is infeasible, so the constraint binds. The KKT equations give

$$\lambda^* = 2, \quad (x^*, y^*) = (0, 1).$$

Example. For

$$\min_{x,y} (x-2)^2 + (y+1)^2 \quad \text{s.t.} \quad x - y \leq 1,$$

the unconstrained point $(2, -1)$ is infeasible. With multiplier λ , stationarity gives $x = 2 - \lambda/2$ and $y = -1 + \lambda/2$. Binding feasibility then gives

$$\lambda^* = 2, \quad (x^*, y^*) = (1, 0).$$

Proposition 6.2: KKT sufficiency for convex problems

Assume that f and every g_i are convex, every equality constraint is affine, and a constraint qualification holds. Then every point satisfying the KKT conditions is a global optimum.

7 Portfolio Optimization

Let $R \in \mathbb{R}^d$ be the vector of asset returns, with

$$\mu = \mathbb{E}[R], \quad \Sigma = \text{Cov}(R).$$

For portfolio weights w ,

$$\mathbb{E}[w^\top R] = \mu^\top w, \quad \text{Var}(w^\top R) = w^\top \Sigma w.$$

Since covariance matrices are positive semidefinite, portfolio variance is a convex quadratic function.

Throughout this section,

$$\mathbf{1} = (1, \dots, 1)^\top \in \mathbb{R}^d$$

denotes the vector of ones, so $\mathbf{1}^\top w = 1$ is the full-investment constraint.

7.1 Minimum variance

Proposition 7.1: Fully invested minimum-variance portfolio

Assume $\Sigma \succ 0$. The problem

$$\min_w \frac{1}{2} w^\top \Sigma w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1$$

has the unique solution

$$w^* = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}.$$

Indeed, stationarity gives $\Sigma w + \lambda \mathbf{1} = 0$, and the budget constraint determines the multiplier.

Example. For $\Sigma = \text{diag}(0.04, 0.09, 0.16)$, inverse-variance weighting gives

$$w^* = \begin{pmatrix} 36/61 \\ 16/61 \\ 9/61 \end{pmatrix}.$$

For the companion covariance $\Sigma = \text{diag}(0.01, 0.04, 0.09)$, the inverse variances are 100, 25, and 100/9, hence

$$w^* = \begin{pmatrix} 36/49 \\ 9/49 \\ 4/49 \end{pmatrix}.$$

In both cases the least volatile asset receives the largest weight.

7.2 Mean–variance trade-off

A common formulation is

$$\min_w \frac{1}{2} w^\top \Sigma w - \gamma \mu^\top w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1.$$

The parameter $\gamma > 0$ controls the reward assigned to expected return. The first-order system is

$$\Sigma w - \gamma \mu + \lambda \mathbf{1} = 0, \quad \mathbf{1}^\top w = 1.$$

Fixing a target return m gives

$$\min_w \frac{1}{2} w^\top \Sigma w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad \mu^\top w = m.$$

Let

$$C = \begin{pmatrix} \mathbf{1}^\top \\ \mu^\top \end{pmatrix}, \quad d = \begin{pmatrix} 1 \\ m \end{pmatrix}.$$

Proposition 7.2: Equality-constrained efficient portfolio

If the required inverses exist, the target-return portfolio is

$$w^* = \Sigma^{-1} C^\top (C \Sigma^{-1} C^\top)^{-1} d.$$

Varying m traces the mean–variance frontier.

7.3 Long-only and operational constraints

With $w \geq 0$, the Lagrangian includes multipliers $\eta \geq 0$ and KKT gives

$$\Sigma w - \gamma \mu + \lambda \mathbf{1} - \eta = 0, \quad \eta_i w_i = 0.$$

Because the problem is convex, KKT conditions are sufficient under the usual constraint qualification.

Real portfolio models may also impose position bounds, sector limits, turnover constraints, transaction costs, tracking-error limits, and stress constraints. Quadratic risk combined with linear constraints remains a convex quadratic program and can be solved reliably at realistic scale.

Constrained rebalancing as a convex QCQP

Consider

$$\min_w \frac{1}{2} w^\top \Sigma w - \gamma \mu^\top w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad w \geq 0, \quad \|w - w^{\text{old}}\|_2^2 \leq \tau^2.$$

Using multipliers $\lambda \in \mathbb{R}$, $\eta \geq 0$, and $\rho \geq 0$, stationarity is

$$\Sigma w - \gamma \mu + \lambda \mathbf{1} - \eta + 2\rho(w - w^{\text{old}}) = 0.$$

Complementary slackness requires

$$\eta_i w_i = 0 \quad \forall i, \quad \rho(\|w - w^{\text{old}}\|_2^2 - \tau^2) = 0.$$

Together with primal and dual feasibility, these are the KKT conditions. When $\Sigma \succeq 0$, the problem is a convex quadratically constrained quadratic program. Under the usual qualification, KKT is sufficient and a convex QCQP solver returns the global optimum.

8 Summary

The workflow is to identify the decision variable, objective, and feasible set; check differentiability and convexity; derive and classify the first-order conditions; state the sign convention and all KKT conditions when constraints are present; choose an algorithm adapted to the structure; verify feasibility, stationarity, and numerical convergence; and interpret the optimizer in the original application.